

INVARIANCE RESEARCH

Agentic Systematic Trading Firm

Product Requirements Document

The 52-Agent Architecture for Continuous Discovery, Falsification, Portfolio Construction, Deployment, and Learning

North-star mandate

Build a closed research institution that continuously discovers predictive relationships, attempts to falsify them, converts survivors into portfolio-ready signals, validates implementation prospectively, deploys under deterministic risk controls, measures live degradation, and compounds institutional knowledge.

Document owner: Founder / Chief Investment Officer

Product scope: Bulletproof BT, Hermes Research Swarm, Research Intelligence Service, Autonomous Alpha
Discovery Laboratory, Portfolio and Production Control Plane

Status: Foundational PRD for phased implementation

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Document Control

Purpose	Define the company-wide agent architecture, operating loop, discovery modes, governance, system requirements, interfaces, permissions, deployment topology, and phased build plan.
Primary users	Founder/CIO, future human team members, Hermes agents, quantitative researchers, developers, risk operators, data engineers, and auditors.
Binding effect	This document defines the target operating model. Individual implementations may evolve, but no component may bypass the promotion state machine, trial registry, independent validation, or deterministic risk controls.
Supersedes	Standalone strategy-tester mentality and any agent design that allows one agent to propose, test, approve, deploy, and judge its own work.
Review cadence	Quarterly architecture review; immediate review after any material live incident, major data-source change, new asset class, or modification to promotion criteria.

Executive Summary

Invariance Research is being designed as a systematic trading firm whose primary competitive advantage is not a single strategy, model, or agent. Its advantage is a governed process for searching more intelligently, falsifying more aggressively, learning cumulatively, and deploying more safely than an informal trading operation.

The product described in this PRD combines Bulletproof BT, a Research Intelligence Service, a 52-profile Hermes swarm, a constrained Autonomous Alpha Discovery Laboratory, portfolio construction services, shadow and micro-live validation, deterministic execution and risk controls, and a durable institutional memory. Together they implement a continuous loop:

```
LEARN -> OBSERVE -> PROPOSE -> REGISTER -> FORMALIZE -> TEST -> FALSIFY
-> COMBINE -> SHADOW -> MICRO-LIVE -> MEASURE DEGRADATION
-> RETAIN / MODIFY / RETIRE -> LEARN AGAIN
```

The system is explicitly not a “smart trader” that is told to find money. It is an autonomous scientific discovery institution. Agents receive almost no credit for spectacular in-sample performance. They are rewarded for producing effects that survive unseen periods, unseen instruments, realistic execution, alternative datasets, perturbations, independent criticism, portfolio tests, prospective observation, and live implementation.

- Bulletproof BT becomes an alpha-research and portfolio-validation platform, not only a strategy backtester.
- Signal discovery is separated from trade-rule construction, portfolio admission, and deployment.
- Every hypothesis, model variation, parameter combination, dataset, code revision, and outcome is registered before results are interpreted.
- Machine learning is used first for conditional trade quality, meta-labelling, regime conditioning, cost and fill prediction, volatility, expected return, and strategy selection - not naive next-bar direction.
- Autonomous search is constrained by a point-in-time-safe research API, a feature/alpha domain-specific language, immutable holdouts, trial-aware statistics, compute budgets, and adversarial evaluation.
- Prop-firm constraints are implemented as a separate portfolio-risk overlay rather than embedded into alpha discovery.
- Production promotion follows a mandatory state machine from proposed to screened, replicated, robust, shadow, micro-live, portfolio-eligible, production, degraded, and retired.

- No Hermes agent can independently increase risk, access hidden holdouts, modify canonical data, alter the execution engine, or deploy unapproved strategies.

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1. Product Vision and Objectives

1.1 Product vision

Create an agentic systematic trading firm that compounds process knowledge. The firm must be able to discover weak but repeatable predictive relationships, distinguish genuine effects from machine-scale false discovery, combine independent signals into portfolios, validate implementation prospectively, deploy through deterministic controls, and learn from the difference between expected and realized behavior.

1.2 Primary objectives

1. Convert Bulletproof BT from a strategy tester into an alpha-research and portfolio-validation platform.
2. Separate signal discovery, strategy construction, portfolio construction, execution, risk, and live learning into governed services and agent roles.
3. Create a shared Research Intelligence Service combining foundational knowledge, current literature, and Invariance Research memory.
4. Enable bounded autonomous discovery over interpretable feature and signal spaces without granting agents unrestricted access to code, data, or holdouts.
5. Introduce machine learning immediately where it improves conditional decisions under costs.
6. Create a trial-aware research process that records failures and adjusts for selection.
7. Build shadow and micro-live stages that validate implementation before material capital is at risk.
8. Maintain a separate prop-firm overlay capable of evaluating pass probability, breach probability, and funded-account survival.
9. Establish strict human and agent decision rights, with deterministic pre-trade and real-time risk enforcement.
10. Measure live degradation and use it to improve research, portfolio, execution, and monitoring methods.

1.3 Non-goals

- Guaranteeing profitable alpha or future returns.
- Building a single omnipotent agent with unrestricted access.
- Using historical profit as the primary reward for autonomous research.
- Training a custom model on every book before a traceable retrieval system exists.
- Allowing online agents to rewrite strategies and trade newly discovered anomalies immediately.
- Using deep reinforcement learning as the first discovery method.
- Optimizing solely for prop-firm challenge rules or one provider's current commercial terms.
- Replacing deterministic order, accounting, reconciliation, or risk controls with free-form language-model reasoning.

2. Operating Principles and Non-Negotiables

Scientific before profitable

The system searches for defensible predictive relationships and survival through falsification gates. Historical profitability is evidence, not proof.

Register before observing

Every hypothesis and candidate specification must be recorded before results are observed. Unregistered results cannot enter the signal library.

Signals before strategies

Research first asks whether information predicts a target. Entry, exit, sizing, and execution wrappers are designed only after predictive content is established.

Broad plateaus over isolated peaks

Stable effects across neighboring parameters, assets, periods, and definitions are preferred over exceptional configurations.

Net economics over model metrics

Net expected value after Tier-2 and Tier-3 cost assumptions is the governing score; accuracy, AUC, and gross Sharpe are secondary diagnostics.

Institutional memory is an asset

Negative results, rejected ideas, failed replications, and live degradation are retained and retrievable.

Independent judgment

Proposers cannot approve their own experiments, validators cannot modify the tested strategy, deployment cannot alter approved logic, and risk can stop but not increase exposure.

Prospective evidence matters

Shadow and micro-live observation are required because historical data cannot fully represent execution, market adaptation, and operational reality.

Complexity pays rent

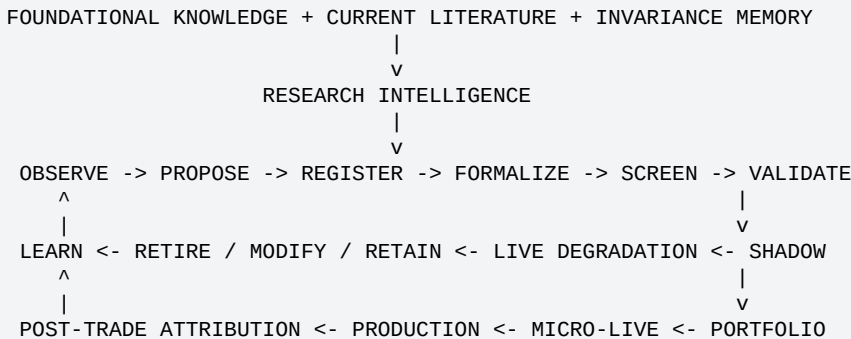
Additional features, model classes, conditions, and parameters must produce measurable incremental value after selection penalties.

Humans remain accountable

Agents prepare, execute, critique, and monitor. Human owners approve research mandates, risk limits, live promotion, capital scaling, and material architectural changes.

3. Systematic Trading Production Loop

3.1 Closed-loop architecture



3.2 Production-loop stages

Stage	Purpose	Mandatory exit artifact
Learn and observe	Retrieve literature, prior experiments, failure patterns, live anomalies, portfolio weaknesses, and execution evidence.	Research brief
Propose	Generate atomic predictive hypotheses with mechanism, target, horizon, expected effect, and falsification conditions.	Hypothesis object
Register	Create immutable hypothesis and experiment records, trial-family counts, data snapshot, code commit, and planned validation.	Registry record
Formalize	Translate the hypothesis into a machine-readable experiment specification using approved features and targets.	Experiment specification
Screen	Run low-cost diagnostic tests designed to reject most candidates quickly.	Screening report
Validate	Use purged walk-forward evaluation, embargo, alternate universes, stress, selection adjustment, replication, and adversarial criticism.	Independent validation package
Construct	Design trade rules only for validated signals; estimate costs, MFE/MAE, holding period, and execution alternatives.	Strategy specification
Combine	Evaluate marginal contribution, correlation, capacity, turnover, and risk within the signal and strategy library.	Portfolio contribution report
Shadow	Generate live predictions and hypothetical orders without capital. Measure data, timing, costs, fills, and forecast calibration.	Prospective validation report
Micro-live	Use minimal capital to validate order state, reconciliation, risk controls, actual costs, and implementation parity.	Implementation validation report
Production and scale	Allocate only after live degradation is measured and acceptable. Scaling is conditional and reversible.	Capital allocation decision
Learn again	Feed outcomes, failures, drift, incidents, and live degradation back into the Research Intelligence Service and trial registry.	Updated institutional memory

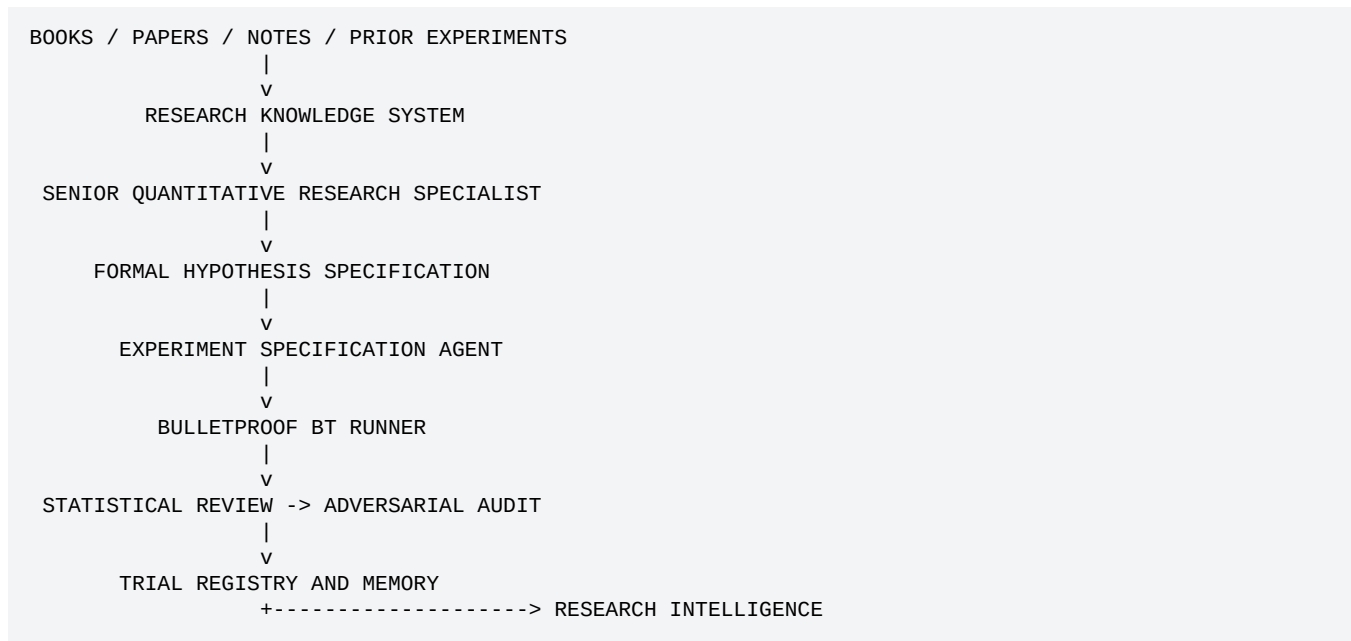
3.3 Feedback loops

- Research feedback: failed and successful results update similar-hypothesis retrieval and method recommendations.

- Execution feedback: predicted versus realized costs update cost models and trade gating.
- Portfolio feedback: correlation, concentration, and marginal contribution shape future research priorities.
- Risk feedback: drawdowns, breaches, and scenario failures alter limits and candidate acceptance criteria.
- Production feedback: live degradation separates research error, implementation error, regime change, and data error.
- Organizational feedback: agent-quality metrics alter prompts, permissions, review requirements, and task routing.

4. Research Intelligence Layer

4.1 Position in the pipeline



The Research Intelligence Layer is built before the entire production swarm is complete. It sits between research data and experiment execution. It does not begin as a smart trader. It begins as a research reasoning system that can retrieve knowledge, inspect prior work, propose mathematically defensible hypotheses, and generate structured specifications for Bulletproof BT.

4.2 Five intelligence types

Domain knowledge

Probability, mathematical statistics, stochastic processes, time-series analysis, econometrics, optimization, information theory, machine learning, market microstructure, behavioral finance, portfolio construction, execution, transaction costs, risk, derivatives, futures, and empirical asset pricing.

Institutional memory

Every hypothesis, trial, dataset, parameter grid, failure, selected configuration, robustness result, shadow observation, live degradation report, note, and methodological decision.

Mathematical reasoning

Methods, dependence, non-stationarity, estimators, assumptions, nulls, sample size, leakage, multiple testing, economic significance, and falsification.

Market context

Instrument, horizon, liquidity, fees, slippage, funding, volatility, exchange constraints, latency, leverage, and liquidation mechanics.

Scientific discipline

Every proposal must state rationale, target, features, data requirements, null, expected effect, failure conditions, validation, cost assumptions, selection adjustment, and deployment implications.

4.3 Retrieval architecture

```
SOURCE DOCUMENTS -> PARSER -> CHUNKING + METADATA -> HYBRID INDEX
-> RESEARCH RETRIEVAL SERVICE -> GENERAL REASONING MODEL
-> STRUCTURED, CITED RESEARCH OUTPUT
```

The initial implementation uses retrieval-augmented generation rather than attempting to train a model to memorize all books and papers. Every material research claim must remain traceable to source, page, section, experiment record, or explicit agent inference.

4.4 Knowledge collections

- Foundational Knowledge: mathematics, statistics, econometrics, machine learning, optimization, finance, and market microstructure.
- Current Research Literature: academic papers, working papers, practitioner research, conference publications, and exchange documentation.
- Invariance Research Memory: Bulletproof BT experiments, notebooks, hypotheses, rejected ideas, strategy specifications, validation reports, cost studies, shadow results, and live degradation.

4.5 Research Intelligence Service APIs

```
POST /v1/research/search
GET /v1/research/sources/{source_id}
POST /v1/research/similar-hypotheses
POST /v1/research/methods/recommend
POST /v1/research/experiments/search
POST /v1/research/hypotheses
POST /v1/research/reviews
POST /v1/research/conclusions
```

4.6 Evidence classification

- Established mathematical result
- Empirical finding
- Author interpretation
- Market-specific evidence
- Invariance Research evidence
- Agent inference

Every conclusion must identify its evidence category. Retrieved material is not automatically true, applicable, current, replicable, or economically meaningful.

5. Autonomous Alpha Discovery Laboratory

5.1 Correct mandate

Prohibited objective

Find a statistical pattern that will surely give us an edge.

Required objective

Discover predictive relationships that survive unseen markets, unseen periods, realistic execution, alternative datasets, perturbations, competing hypotheses, prospective observation, and trial-aware statistical scrutiny.

An agent rewarded for finding a sure edge may exploit leakage, timestamps, simulator flaws, excessive search, lucky holdouts, hidden tail risk, or unrealistic fills. The laboratory must therefore reward survival through falsification, not historical profit alone.

5.2 Discovery score

$$\text{Discovery Score} = P(\text{effect is real}) \times P(\text{effect persists}) \times \text{economic magnitude} \times \text{capacity} \\ - \text{cost} - \text{complexity} - \text{selection penalty}$$

$$R = w1*OOS_IC + w2*OOS_EV + w3*Stability + w4*CrossMarket \\ + w5*CostSurvival - w6*Drawdown - w7*Turnover - w8*Complexity \\ - w9*TrialPenalty - w10*RegimeConcentration$$

5.3 Market laboratory

- Versioned OHLCV, trades, order books where available, funding, open interest, liquidations, basis, spreads, depth, cross-asset prices, exchange state, time variables, liquidity and volatility regimes, benchmark assets, and known protocol or corporate events.
- Point-in-time-safe API access rather than unrestricted database access.
- Delayed availability, survivorship and delisting handling, immutable partitions, query logging, feature lineage, and sealed holdouts.
- No direct agent access to hidden validation files.

5.4 Constrained alpha language

Initial autonomous discovery uses a constrained symbolic language rather than arbitrary Python. The DSL supports lags, differences, returns, rolling statistics, ranks, z-scores, correlations, residuals, quantiles, entropy, acceleration, cross-sectional dispersion, volatility scaling, interactions, conditional operators, neutralization, and regime gates.

```
signal:
  expression:
    multiply:
      - rank:
          input: {rolling_zscore: {input: funding_rate, window: 72}}
      - negate:
          rank:
            input: {return: {horizon: 12}}
universe: {liquidity_rank_max: 30}
holding_period: {bars: 6}
neutralization: {market_beta: true, volatility: true}
```

5.5 Search scheduler

11. Test a large candidate set using inexpensive screening.
12. Retain a small fraction using predeclared filters.
13. Expand periods, instruments, costs, and controls.
14. Run full walk-forward and trial-aware evaluation.
15. Adversarially test the survivors.
16. Shadow only a small number of candidates.
17. Promote to micro-live only after implementation readiness.

5.6 Research self-play analogues

Researcher versus Critic

Explorer earns reward when a signal survives independent validation. Critic earns reward for exposing leakage, instability, execution flaws, or hidden concentration.

Strategy versus Market Adversary

A stress agent changes plausible spreads, latency, volatility, liquidity, correlation, funding, gaps, and fill probability to expose fragility.

Current model versus archived models

A synthetic league tests policies against mixtures of earlier policies and participant types. It is a robustness laboratory, not proof of real-market profitability.

Researcher versus Red Team

The Red Team attempts to infer holdouts, exploit caches, alter costs, submit invalid values, exploit bar boundaries, or overwrite outputs. The system must detect and block these behaviors.

6. Discovery Modes

6.1 Mode A - Literature-led hypothesis discovery

The Senior Quantitative Research Specialist retrieves mathematical and empirical literature, compares it with prior Invariance evidence, and proposes a pre-registered atomic hypothesis.

6.2 Mode B - Institutional-memory-led discovery

The system mines failures, partial successes, live degradation, execution errors, and unresolved research questions to identify nonredundant follow-up studies.

6.3 Mode C - Human-observation-led discovery

A founder or researcher records a market observation. The system converts it into a falsifiable claim and searches for related prior work before any test runs.

6.4 Mode D - Autonomous factor mining

Symbolic regression, genetic programming, Bayesian optimization, random grammar search, gradient-guided factor generation, and LLM-guided composition search interpretable candidate factors within the DSL.

6.5 Mode E - Cross-sectional and cross-market discovery

Agents search rankings, residual returns, relative-value relationships, cross-exchange lead-lag, carry, basis, funding, dispersion, and cross-asset conditioning. Replication across markets helps distinguish universal mechanisms from crypto-specific artifacts.

6.6 Mode F - Event and microstructure discovery

Agents study liquidations, order-flow pressure, spread behavior, funding resets, volatility shocks, depth changes, abnormal volume, and exchange-specific events.

6.7 Mode G - ML-assisted conditional discovery

ML identifies when a known signal is likely to work, fail, become expensive, or become dangerous. The output is conditional trade quality, expected return, risk, cost, or survivability - not generic directional prediction.

6.8 Mode H - Researcher-critic adversarial league

Explorer, experiment designer, statistical auditor, falsifier, and research governor form a bounded scientific game. The objective is fast rejection of false edges and durable retention of survivors.

6.9 Mode I - Synthetic multi-agent market laboratory

Noise traders, market makers, trend followers, mean reverters, liquidators, arbitrageurs, informed agents, and archived policies create plausible counterfactual environments. Results remain hypotheses until validated against historical and prospective real data.

6.10 Mode J - Prospective shadow discovery

Live observation can identify anomalies and distribution shifts, but newly discovered effects cannot be traded immediately. They return to the research pipeline for registration, testing, and governance.

7. Machine Learning Product Roadmap

7.1 Assumptions to kill

Incorrect assumption	Required replacement
ML predicts returns.	ML predicts conditional trade quality, risk, costs, fill probability, expected return, and survivability under costs.
More features help.	Every feature must be timestamp-provable, economically interpretable or empirically justified, versioned, and leakage-audited.
Offline accuracy matters.	Net expected value under realistic Tier-2 and Tier-3 assumptions is the governing score.
Large row counts mean large samples.	Effective sample size must account for overlap, common exposure, dependence, regime concentration, and repeated events.
A more complex model is better.	The complex model must beat transparent baselines after costs, uncertainty, and selection penalties.

7.2 Reality-aligned labels

- Barrier/event label: Did the opportunity reach +aR before -bR within horizon H?

- Every label stores label_end_time to enable purging and embargo.
- Additional targets: forward residual return, volatility, MFE, MAE, fill probability, slippage, continuation probability, and drawdown contribution.
- Labels must match the actual decision the model will control.

7.3 Initial decision roles

- Enter or skip
- Size up or down within deterministic limits
- Select among approved strategies
- Condition a signal on regime
- Estimate cost and fill quality
- Estimate expected return and uncertainty
- Forecast volatility and correlation

7.4 Model sequence

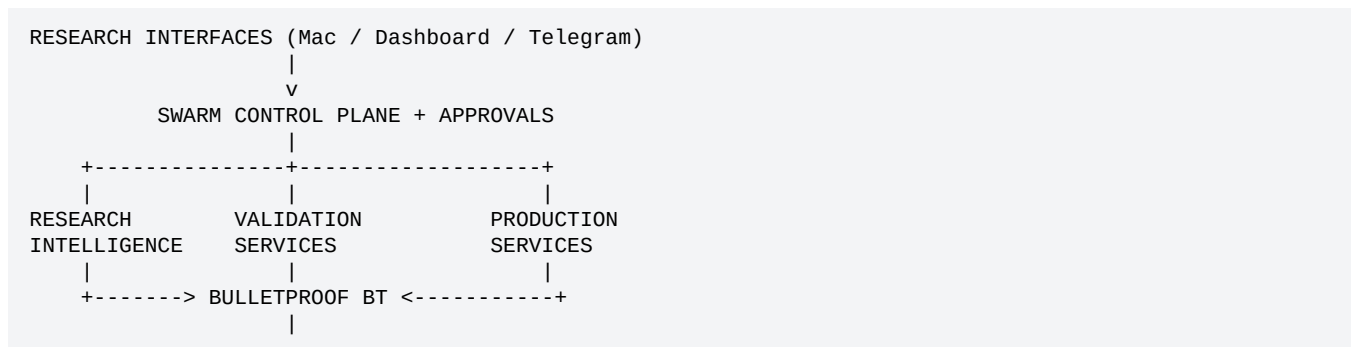
18. Create transparent baselines: logistic regression, regularized linear models, simple thresholds, and calibrated scorecards.
19. Add gradient boosting, random forests, generalized additive models, and other moderate-complexity models.
20. Use neural methods only where sample structure, data quality, and incremental value justify them.
21. Use offline reinforcement learning downstream of a validated signal library for act/skip, allocation, inventory, combination, and exit decisions.
22. Use synthetic multi-agent environments for robustness learning, never as sole evidence of real-market profitability.

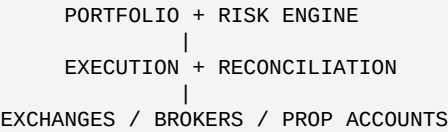
7.5 ML acceptance criteria

- Improves net out-of-sample expected value, risk, cost, or portfolio contribution versus the approved baseline.
- Remains calibrated prospectively.
- Survives purged walk-forward validation, embargo, drift tests, and alternate samples.
- Has documented feature availability and model lineage.
- Fails safely when inputs are missing, stale, shifted, or outside training support.
- Can be paused or replaced without disabling deterministic risk controls.

8. Product Architecture and Core Services

8.1 Logical architecture





Shared foundations: PostgreSQL/pgvector, artifact store, logs, metrics, secrets, versioned data

8.2 Required core services

Service	Requirement
Research Intelligence Service	Hybrid retrieval, citations, methods, prior experiments, similar hypotheses, conclusions.
Research Registry	Hypotheses, experiments, trials, results, relationships, failures, and decisions.
Market Research API	Point-in-time-safe data access with logged queries and immutable snapshots.
Feature and Alpha DSL Compiler	Validates symbolic expressions and compiles them into causal feature pipelines.
Experiment Orchestrator	Runs registered jobs, enforces manifests, compute budgets, retries, and artifact capture.
Statistical Evaluation Service	Purging, embargo, bootstrap, DSR, PBO, multiple-testing correction, effective sample size, and stability.
Adversarial Validation Service	Stress, placebo, alternate universe, timestamp shift, outlier removal, and simulator-exploit tests.
Signal Library	Versioned approved signals, forecast calibration, status, lineage, and expected scope.
Portfolio Construction Service	Forecast normalization, allocation, constraints, turnover, capacity, and marginal contribution.
Prop-Firm Overlay Service	Firm-specific account rules, breach simulation, pass probability, and survival analytics.
Shadow Market Service	Live features, predictions, hypothetical orders, observable fills, latency, and prospective outcomes.
Execution and OMS	Deterministic order generation, state management, duplicate prevention, retries, and acknowledgements.
Pre-Trade and Real-Time Risk	Deterministic limits, stale-data controls, exposure, drawdown, halts, and kill switches.
Reconciliation Service	Independent comparison of internal and external positions, orders, balances, and fills.
Observability and Incident Service	Logs, metrics, traces, alerts, incidents, runbooks, and postmortems.
Artifact and Lineage Service	Immutable references to data, code, configuration, reports, models, containers, and decisions.

9. The 52-Agent Operating Model

The architecture contains exactly 52 governed agent profiles. They are reusable specialist roles, not necessarily 52 continuously running processes. The control plane instantiates profiles per task, applies least privilege, records every action, and routes outputs to independent reviewers.

Senior research specialist profile

The principal research-intelligence agent is intentionally specified as a computer-native polymath with doctoral-level competence across statistics, mathematics, physics, computer science, and econometrics simultaneously, plus time-series

research, ML validation, portfolio construction, market microstructure, independent research, scientific writing, and reproducible computation. This describes the capability envelope expected from the agent, not a realistic requirement for one human hire.

1. Research Program Director Agent

Division: Executive Coordination

Coordinates the research agenda and converts firm priorities into bounded programs.

Core responsibilities	Maintain backlog; group hypotheses; allocate compute; prevent duplication; track dependencies; surface portfolio gaps.
Required outputs	Program charter; ranked backlog; weekly plan; budget recommendation; escalation report.
Permissions and prohibitions	May create and assign research tasks; may not approve promotion, alter live risk, or trade.
Activation triggers	New CIO directive; weekly planning; portfolio gap; failed program.
Key dependencies	Research Registry; Portfolio Analyst; CIO.
Success measures	Research throughput, duplicate work avoided, value of information per compute unit.

2. Investment Committee Secretary Agent

Division: Executive Coordination

Builds complete evidence packages and preserves the decision record.

Core responsibilities	Collect artifacts; verify gates; summarize dissent; create approval memo; track conditions and expiry.
Required outputs	Investment memo; gate checklist; dissent summary; decision record.
Permissions and prohibitions	No authority to approve, deploy, or modify evidence.
Activation triggers	Candidate reaches committee gate; conditional review; model expiry.
Key dependencies	All validation agents; CIO; Risk Officer.
Success measures	Completeness, traceability, zero missing gate artifacts.

3. Trading Operations Coordinator Agent

Division: Executive Coordination

Coordinates approved candidates through shadow, demo, micro-live, production, and retirement.

Core responsibilities	Verify prerequisites; schedule releases; coordinate deployment; track lifecycle; manage incident handoffs.
Required outputs	Release plan; readiness checklist; lifecycle status; incident

	coordination record.
Permissions and prohibitions	Cannot change strategy logic, capital limits, or bypass approvals.
Activation triggers	Promotion decision; scheduled release; production incident.
Key dependencies	Deployment; Risk; Reconciliation; Observability.
Success measures	Release reliability, rollback readiness, incident resolution quality.

4. Senior Quantitative Research Specialist Agent

Division: Research Intelligence

Acts as the firm's principal quantitative research strategist and polymath reasoning system.

Core responsibilities	Synthesize statistics, mathematics, physics, CS, econometrics, stochastic processes, time series, ML, portfolio construction, market microstructure, execution, and prior Invariance evidence; propose falsifiable hypotheses; identify methods, assumptions, sample requirements, failure modes, and nonredundant directions; review independent research quality.
Required outputs	Cited research brief; formal hypothesis object; method recommendation; prior-art map; assumptions register; research critique.
Permissions and prohibitions	Read-only access to approved knowledge and experiment records; cannot run arbitrary code, see sealed holdouts, approve promotion, deploy, or trade.
Activation triggers	Research objective; unresolved anomaly; failed replication; portfolio gap; literature update.
Key dependencies	Research Intelligence Service; Hypothesis Architect; Experiment Designer; Trial Registry.
Success measures	Novel but defensible hypotheses, low redundancy, accurate source use, high falsification yield, cumulative learning.

5. Market Literature Research Agent

Division: Research Intelligence

Retrieves and summarizes relevant academic and practitioner evidence.

Core responsibilities	Search sources; extract methods; compare claims; identify applicability and limitations; preserve citations.
Required outputs	Literature dossier; evidence table; source passages; replication notes.
Permissions and prohibitions	No unsupported claims; cannot convert papers directly into approved strategies.
Activation triggers	Research question; requested method review; new source

	ingestion.
Key dependencies	Research Intelligence Service; Senior Specialist.
Success measures	Citation accuracy, relevance, diversity, and reproducibility.

6. Crypto Market Structure Agent

Division: Research Intelligence

Generates domain-specific hypotheses from crypto mechanics.

Core responsibilities	Study funding, basis, liquidations, open interest, exchange fragmentation, stablecoin flows, leverage, token events, and venue behavior.
Required outputs	Market-structure memo; candidate mechanisms; data requirements; event taxonomy.
Permissions and prohibitions	No execution or exchange-account permissions.
Activation triggers	New market event; research program; observed anomaly.
Key dependencies	Market API; Alternative Data Scout; Event Researcher.
Success measures	Unique crypto-specific hypotheses that survive basic plausibility review.

7. Cross-Asset Research Agent

Division: Research Intelligence

Tests whether effects are universal, asset-class-specific, or hidden beta.

Core responsibilities	Compare crypto, equity index futures, FX, rates, commodities, and benchmarks; propose replication markets.
Required outputs	Cross-market comparison; replication plan; hidden-exposure assessment.
Permissions and prohibitions	No strategy promotion authority.
Activation triggers	Candidate effect; benchmark failure; portfolio concentration.
Key dependencies	Market API; Benchmark Agent; Portfolio Agent.
Success measures	Replication quality and reduction of market-specific false conclusions.

8. Alternative Data Scout Agent

Division: Research Intelligence

Evaluates datasets before acquisition or use.

Core responsibilities	Assess coverage, latency, licensing, quality, history, survivorship, point-in-time properties, cost, and expected research value.
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Required outputs	Data due-diligence memo; acquisition priority; schema proposal.
Permissions and prohibitions	Cannot purchase data or ingest unapproved sources.
Activation triggers	New research need; vendor proposal; missing feature.
Key dependencies	Data Engineer; Compliance; Research Director.
Success measures	Useful data acquired per cost and avoided low-quality sources.

9. Hypothesis Architect Agent

Division: Quantitative Research

Converts observations into complete pre-registered hypothesis objects.

Core responsibilities	Define mechanism, target, horizon, universe, expected effect, null, failure conditions, cost sensitivity, and related trials.
Required outputs	Machine-readable hypothesis specification.
Permissions and prohibitions	Cannot modify hypothesis after seeing results without creating a new version.
Activation triggers	Approved research brief or human observation.
Key dependencies	Senior Specialist; Trial Registry.
Success measures	Completeness, falsifiability, and low post-hoc modification rate.

10. Experiment Specification Agent

Division: Quantitative Research

Transforms hypotheses into executable, machine-readable designs.

Core responsibilities	Define features, labels, datasets, splits, purging, embargo, controls, costs, stress, trial budget, and success criteria.
Required outputs	Experiment manifest and validation plan.
Permissions and prohibitions	Cannot view results before specification is locked; cannot choose favorable tests post hoc.
Activation triggers	Registered hypothesis approved for testing.
Key dependencies	Feature Registry; Data API; Statistical Reviewer.
Success measures	Executable specifications with no unresolved ambiguity or leakage.

11. Signal Researcher Agent

Division: Quantitative Research

Determines whether a variable contains predictive information before trade rules are added.

Core responsibilities	Forward-return surfaces; deciles; IC; monotonicity; horizons; regime and asset breadth; gross/net feasibility.
Required outputs	Signal evidence report; candidate signal object.
Permissions and prohibitions	No full strategy optimization during signal stage.
Activation triggers	Approved experiment manifest.
Key dependencies	Executor; Statistical Service.
Success measures	Stable, interpretable predictive relationships and high early rejection rate.

12. Strategy Construction Agent

Division: Quantitative Research

Builds tradeable wrappers around validated signals.

Core responsibilities	Entry, exit, holding, stops, execution type, throttling, risk unit, and operational constraints.
Required outputs	Versioned strategy specification.
Permissions and prohibitions	Cannot alter underlying signal evidence or validation splits.
Activation triggers	Signal passes discovery gate.
Key dependencies	Execution Cost Agent; MFE/MAE models; Portfolio Agent.
Success measures	Net monetization of signal without fragile parameterization.

13. Cross-Sectional Research Agent

Division: Quantitative Research

Researches rankings, residual returns, spreads, and relative-value portfolios.

Core responsibilities	Rank assets; neutralize beta; test cross-sectional persistence/reversal; control sectors and liquidity.
Required outputs	Factor study; cross-sectional signal; neutralization specification.
Permissions and prohibitions	No direct order placement.
Activation triggers	Cross-sectional research program.
Key dependencies	Market API; Portfolio Agent.
Success measures	Breadth, neutrality, replication, and distinct portfolio contribution.

14. Time-Series Research Agent

Division: Quantitative Research

Researches temporal effects within instruments.

Core responsibilities	Trend, carry, breakout, reversal, volatility conditioning, persistence, and decay across horizons.
Required outputs	Time-series signal report and candidate definitions.
Permissions and prohibitions	No parameter winner selection without plateau analysis.
Activation triggers	Time-series program.
Key dependencies	Signal Researcher; Benchmark Agent.
Success measures	Temporal stability and cost-adjusted significance.

15. Event-Driven Research Agent

Division: Quantitative Research

Researches discrete market and microstructure events.

Core responsibilities	Liquidations, funding resets, announcements, volume shocks, volatility shocks, gaps, and exchange events.
Required outputs	Event taxonomy; event-study report; candidate event signals.
Permissions and prohibitions	Cannot use future event knowledge or unverified timestamps.
Activation triggers	Registered event research program.
Key dependencies	Market Structure Agent; Data Quality.
Success measures	Point-in-time-correct event effects with realistic tradability.

16. Machine Learning Research Agent

Division: Quantitative Research

Develops ML models for conditional decisions and forecasts.

Core responsibilities	Baselines; feature selection; calibration; purged validation; meta-labelling; expected return, risk, cost, and fill models.
Required outputs	Model card; training manifest; validation report; inference contract.
Permissions and prohibitions	Cannot train on sealed holdouts, optimize classification accuracy alone, or deploy itself.
Activation triggers	Validated target and approved feature set.
Key dependencies	Feature Agent; Statistical Reviewer; Model Risk.
Success measures	Net incremental value, calibration, stability, and safe failure.

17. Regime Modelling Agent

Division: Quantitative Research

Creates and validates market-state representations.

Core responsibilities	GMM, HMM, change points, clustering, transition analysis, and downstream decision tests.
Required outputs	Regime model card; state definitions; stability report.
Permissions and prohibitions	Clusters are not accepted without downstream benefit.
Activation triggers	Regime research program or model instability.
Key dependencies	ML Agent; Portfolio; Risk Forecasting.
Success measures	Improved decisions, stable states, and limited regime concentration.

18. Feature Engineering Agent

Division: Quantitative Research

Creates causal, reusable, versioned features.

Core responsibilities	Define formulas, availability timestamps, lags, normalization, missing policy, transformations, redundancy, and lineage.
Required outputs	Feature definition; tests; version; documentation.
Permissions and prohibitions	Cannot silently change feature semantics or backfill unavailable data.
Activation triggers	New experiment or model feature request.
Key dependencies	Market API; Leakage Auditor.
Success measures	Timestamp correctness, reuse, low redundancy, and batch/stream parity.

19. Autonomous Factor Explorer Agent

Division: Autonomous Discovery

Searches the constrained DSL for atomic predictive statements.

Core responsibilities	Compose factors; propose mechanisms; query prior failures; generate registered candidates within budgets.
Required outputs	Candidate hypothesis and DSL expression.
Permissions and prohibitions	No arbitrary Python, hidden data, strategy deployment, or unregistered trials.
Activation triggers	Bounded autonomous discovery session.
Key dependencies	Research Governor; DSL Compiler; Trial Registry.
Success measures	Useful novelty per trial, interpretability, and low rule violations.

20. Symbolic Search Scheduler Agent

Division: Autonomous Discovery

Allocates candidate search using successive halving and value-of-information logic.

Core responsibilities	Schedule cheap screens; promote fractions; enforce family budgets; prevent early noise monopolization.
Required outputs	Search plan; budget ledger; candidate funnel report.
Permissions and prohibitions	Cannot change validation criteria or hide rejected candidates.
Activation triggers	Discovery program approval.
Key dependencies	Explorer; Compute Scheduler; Trial Registry.
Success measures	Compute efficiency and calibrated survivor rates.

21. Market Adversary Agent

Division: Autonomous Discovery

Generates plausible hostile execution and market conditions.

Core responsibilities	Perturb spreads, fees, latency, volatility, liquidity, correlation, funding, gaps, and fill probabilities.
Required outputs	Adversarial scenario set and fragility report.
Permissions and prohibitions	Cannot modify canonical results or choose impossible scenarios without labeling them.
Activation triggers	Candidate enters robustness stage.
Key dependencies	Stress Agent; Risk Forecasting.
Success measures	Fragilities found before deployment and plausible scenario coverage.

22. Synthetic Market League Agent

Division: Autonomous Discovery

Runs candidate policies against archived and synthetic participant mixtures.

Core responsibilities	Maintain archived policies; simulate market makers, noise, trend, reversion, liquidators, arbitrageurs, informed agents.
Required outputs	League results; robustness diagnostics; exploitability report.
Permissions and prohibitions	Results cannot count as real-market validation.
Activation triggers	Approved synthetic research program.
Key dependencies	Agent-based simulator; Research Governor.
Success measures	Behavioral robustness and reduced single-opponent exploitation.

23. Laboratory Red Team Agent

Division: Autonomous Discovery

Attempts to exploit the research environment itself.

Core responsibilities	Probe holdouts, filenames, caches, cost settings, NaNs, bar boundaries, permissions, output overwrites, and query side channels.
Required outputs	Security finding; exploit reproduction; remediation test.
Permissions and prohibitions	No production secrets or destructive actions beyond isolated sandbox.
Activation triggers	Scheduled red-team cycle or architecture change.
Key dependencies	Security Agent; Research Governor.
Success measures	Cheating paths detected and closed.

24. Live Anomaly Scout Agent

Division: Autonomous Discovery

Observes live data for unmodeled behavior without trading new discoveries.

Core responsibilities	Detect drift, unusual relationships, unexplained cost changes, and candidate anomalies; create research observations.
Required outputs	Anomaly record; proposed research question; supporting live evidence.
Permissions and prohibitions	Cannot rewrite or trade strategies; observations must re-enter registration.
Activation triggers	Fast-clock live observation.
Key dependencies	Shadow Service; Research Registry.
Success measures	Useful anomalies, low false-alarm burden, zero unauthorized adaptation.

25. Research Governor Agent

Division: Autonomous Discovery

Controls objectives, permissions, budgets, holdouts, and promotion gates.

Core responsibilities	Issue bounded mandates; enforce trial limits; validate allowed tools; stop violations; route approvals.
Required outputs	Governance record; budget decision; gate result.
Permissions and prohibitions	Cannot invent alpha, modify evidence, or approve capital alone.
Activation triggers	Every autonomous session and promotion request.
Key dependencies	Control Plane; Human approvers; Security.

Success measures	No gate bypasses, budget compliance, and complete audit trails.
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26. Research Execution Agent

Division: Autonomous Discovery

Executes approved manifests through Bulletproof BT and auxiliary tools.

Core responsibilities	Compile DSL; launch jobs; preserve configs; collect outputs; report failures; never alter hypothesis post-result.
Required outputs	Run artifacts; logs; checksums; status.
Permissions and prohibitions	Cannot modify data, engine, fee models, splits, or experiment records.
Activation triggers	Approved experiment manifest.
Key dependencies	Backtest Orchestrator; Artifact Store.
Success measures	Reproducibility, job integrity, and low execution error.

27. Statistical Reviewer Agent

Division: Independent Validation

Independently reproduces results in a clean workspace and evaluates evidence with trial-aware statistical methods.

Core responsibilities	Clean-workspace reproduction; code/data/config verification; effect size; confidence; bootstrap; dependence; effective sample; DSR; PBO; FDR; stability; economic significance.
Required outputs	Statistical validation report and confidence grade.
Permissions and prohibitions	Cannot redesign hypothesis after results or approve live capital.
Activation triggers	Completed experiment set.
Key dependencies	Trial Registry; Statistical Service.
Success measures	Calibration of judgments and reduction of false positives.

28. Leakage and Bias Auditor Agent

Division: Independent Validation

Detects lookahead, timestamp, universe, survivorship, overlap, and data leakage.

Core responsibilities	Audit availability; labels; splits; delistings; duplicates; feature lineage; execution alignment.
Required outputs	Leakage report; severity; remediation requirement.
Permissions and prohibitions	Can block promotion; cannot repair silently.

Activation triggers	Every candidate validation and data change.
Key dependencies	Data Quality; Feature Registry; Market API.
Success measures	Leakage caught before promotion and zero unresolved critical findings.

29. Adversarial Research Auditor Agent

Division: Independent Validation

Attempts to disprove the economic and empirical claim.

Core responsibilities	Alternative explanations; hidden beta; one-period dependence; benchmark attacks; mechanism challenges; counterfactual tests.
Required outputs	Adversarial review; competing hypotheses; kill recommendation.
Permissions and prohibitions	Cannot modify candidate or selectively omit favorable evidence.
Activation triggers	Candidate passes initial statistics.
Key dependencies	Benchmark Agent; Cross-Asset Agent.
Success measures	Weak candidates killed early and useful competing explanations.

30. Robustness and Stress Agent

Division: Independent Validation

Tests sensitivity to costs, data, delay, outages, universes, parameters, and structural breaks.

Core responsibilities	Fee/slippage/delay stress; missing data; partial fills; alternate bars; asset/year removal; neighboring parameters.
Required outputs	Robustness matrix and survival grade.
Permissions and prohibitions	Cannot choose only favorable stresses.
Activation triggers	Candidate enters robustness state.
Key dependencies	Market Adversary; Execution Models.
Success measures	Coverage and correlation with future live degradation.

31. Benchmark and Replication Agent

Division: Independent Validation

Compares complexity with simple baselines and external replications.

Core responsibilities	Buy/hold; simple trend; simple reversion; matched random; volatility-scaled; turnover-matched; cross-market replication.
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Required outputs	Benchmark report; incremental-value decision.
Permissions and prohibitions	Cannot approve complexity without demonstrated value.
Activation triggers	Any candidate or ML model.
Key dependencies	Cross-Asset Research; Portfolio.
Success measures	Complexity justified and hidden exposure identified.

32. Forecast Normalization Agent

Division: Portfolio and Risk

Converts heterogeneous signals into comparable, calibrated forecasts.

Core responsibilities	Scale, winsorize, calibrate, uncertainty-discount, and monitor forecast distributions.
Required outputs	Standardized forecast stream and calibration report.
Permissions and prohibitions	Cannot create new alpha or exceed approved ranges.
Activation triggers	Signal admission or periodic recalibration.
Key dependencies	Signal Library; Model Risk.
Success measures	Stable comparable forecasts and prospective calibration.

33. Portfolio Construction Agent

Division: Portfolio and Risk

Combines forecasts under risk, cost, liquidity, and concentration constraints.

Core responsibilities	Optimize allocation; cap exposure; penalize turnover; model covariance; handle constraints and uncertainty.
Required outputs	Target portfolio; constraint report; expected risk and cost.
Permissions and prohibitions	Cannot override hard risk limits or use unapproved signals.
Activation triggers	Portfolio rebalance or candidate admission.
Key dependencies	Risk Forecasting; Allocation; Prop Overlay.
Success measures	Net portfolio efficiency, diversification, and constraint compliance.

34. Strategy Allocation Agent

Division: Portfolio and Risk

Allocates risk across strategy sleeves based on marginal value.

Core responsibilities	Evaluate correlation, drawdown, capacity, confidence, decay, and redundancy; recommend scale/deallocation.
Required outputs	Sleeve allocation recommendation.

Permissions and prohibitions	Cannot allocate solely on recent P&L or increase firm limits.
Activation triggers	Periodic allocation review or candidate admission.
Key dependencies	Portfolio Construction; Performance Analyst.
Success measures	Marginal contribution and controlled concentration.

35. Risk Forecasting Agent

Division: Portfolio and Risk

Forecasts volatility, covariance, tail risk, concentration, and liquidation exposure.

Core responsibilities	Scenario analysis; stress correlations; regime risks; liquidity-adjusted risk; uncertainty ranges.
Required outputs	Risk forecast and scenario pack.
Permissions and prohibitions	No trading or limit increases.
Activation triggers	Pre-allocation, daily risk cycle, regime shift.
Key dependencies	Market Data; Regime Agent.
Success measures	Risk forecast calibration and early warning value.

36. Prop-Firm Constraint Agent

Division: Portfolio and Risk

Implements account-provider rules as a separate portfolio overlay.

Core responsibilities	Daily loss, max/trailing drawdown, leverage, overnight, event, consistency, payout, reset, and breach simulations.
Required outputs	Pass probability; breach probability; time-to-target; survival score; risk multiplier.
Permissions and prohibitions	Cannot redefine alpha or use stale provider rules.
Activation triggers	Provider selection; account change; portfolio review.
Key dependencies	Risk Engine; Operations; current rule configuration.
Success measures	Accurate rule enforcement and funded-account survival.

37. Pre-Trade Risk Agent

Division: Risk Control

Produces deterministic approve, reduce, reject, or halt decisions before orders.

Core responsibilities	Check authorization, exposure, leverage, drawdown, liquidity, stale data, account rules, and duplicate intent.
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Required outputs	Signed risk decision with reasons.
Permissions and prohibitions	Cannot be overridden by execution; LLM commentary is nonbinding.
Activation triggers	Every proposed order.
Key dependencies	Deterministic Risk Engine; Prop Overlay.
Success measures	Zero unauthorized orders and predictable latency.

38. Real-Time Risk Sentinel Agent

Division: Risk Control

Monitors live risk and invokes deterministic protection.

Core responsibilities	Positions, P&L, drawdown, leverage, correlations, data health, abnormal fills, exchange health, and strategy divergence.
Required outputs	Alerts; cancel/reduce/halt instructions; incident record.
Permissions and prohibitions	May reduce or halt; may not increase limits or initiate discretionary trades.
Activation triggers	Continuous live operation.
Key dependencies	Risk Engine; OMS; Observability.
Success measures	Loss containment, fast detection, no false increases.

39. Model Risk Agent

Division: Risk Control

Monitors assumptions, drift, calibration, support, and decay.

Core responsibilities	Feature drift; forecast drift; calibration; performance decay; review dates; model inventory.
Required outputs	Model risk status; pause/review recommendation.
Permissions and prohibitions	Cannot retrain or deploy models independently.
Activation triggers	Scheduled review or threshold breach.
Key dependencies	ML Models; Shadow; Performance.
Success measures	Early decay detection and controlled model lifecycle.

40. Operational Risk Agent

Division: Risk Control

Monitors infrastructure and process failure risks.

Core responsibilities	Connectivity; credentials; duplicate orders; backups; reconciliation; permissions; incident controls.
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Required outputs	Operational risk dashboard; incidents; remediation tracking.
Permissions and prohibitions	May halt affected workflows; cannot alter trading logic.
Activation triggers	Continuous operation and release events.
Key dependencies	Security; Observability; Reconciliation.
Success measures	Reduced operational incidents and rapid containment.

41. Execution Strategy Agent

Division: Trading and Execution

Selects approved execution tactics and venues for target orders.

Core responsibilities	Market/passive choice; urgency; slicing; venue comparison; spread/depth; fees; funding; fill probability; adverse selection; inventory and counterparty awareness.
Required outputs	Execution plan and expected cost.
Permissions and prohibitions	Cannot alter target position or bypass risk.
Activation triggers	Approved target order.
Key dependencies	Cost Model; Market Data; OMS.
Success measures	Implementation shortfall versus benchmark.

42. Order Management Agent

Division: Trading and Execution

Runs deterministic order state management.

Core responsibilities	Create, acknowledge, cancel, replace, retry, handle partial fills, prevent duplicates, and persist states.
Required outputs	Order and event records.
Permissions and prohibitions	No free-form discretionary behavior; cannot override risk or change strategy.
Activation triggers	Approved order.
Key dependencies	Exchange Adapter; Risk Decision.
Success measures	State integrity, low error, no duplicate orders.

43. Execution Quality Analyst Agent

Division: Trading and Execution

Measures predicted versus realized execution.

Core responsibilities	Implementation shortfall; fill ratio; latency; adverse selection; fees; funding; cost-model residuals.
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Required outputs	Execution-quality report and cost-model feedback.
Permissions and prohibitions	Cannot rewrite historical fills or hide bad execution.
Activation triggers	Post-trade and daily review.
Key dependencies	Orders; Fills; Cost Models.
Success measures	Cost prediction accuracy and execution improvement.

44. Position Reconciliation Agent

Division: Trading and Execution

Independently reconciles internal and external records.

Core responsibilities	Positions, balances, open orders, fills, fees, funding, and cash movements.
Required outputs	Reconciliation status and discrepancy incidents.
Permissions and prohibitions	Independent from OMS; may block new trading on critical mismatch.
Activation triggers	Scheduled intervals and post-incident.
Key dependencies	Exchange APIs; Ledger; OMS.
Success measures	Zero unexplained material discrepancies.

45. Shadow Market Agent

Division: Trading and Execution

Runs prospective live observation without capital.

Core responsibilities	Compute registered features; predictions; hypothetical orders; expected and observable fills; latency; forward returns; divergence.
Required outputs	Shadow dataset and prospective validation report.
Permissions and prohibitions	Cannot place live orders or modify research definitions.
Activation triggers	Candidate enters shadow state.
Key dependencies	Live Data; Signal Service; Cost Models.
Success measures	Prospective calibration, implementation realism, and data parity.

46. Market Data Ingestion Agent

Division: Data and Platform

Ingests raw and reference market data reliably.

Core responsibilities	Trades, bars, books, funding, open interest, liquidations,
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	instrument metadata, and exchange status.
Required outputs	Versioned raw datasets; ingestion logs; checksums.
Permissions and prohibitions	Cannot silently transform or impute raw data.
Activation triggers	Continuous feeds or backfill job.
Key dependencies	Vendor/Exchange APIs; Storage.
Success measures	Completeness, timeliness, and exact replayability.

47. Data Quality and Curation Agent

Division: Data and Platform

Detects defects and produces approved curated snapshots.

Core responsibilities	Missing/duplicate/stale timestamps; symbol maps; universe history; delistings; anomalies; quarantine; manifests.
Required outputs	Quality report; curated snapshot; manifest; quarantine record.
Permissions and prohibitions	May quarantine; may not silently repair or overwrite raw data.
Activation triggers	New data arrival, snapshot creation, or anomaly.
Key dependencies	Ingestion; Research Registry.
Success measures	Defect detection, point-in-time correctness, and reproducibility.

48. Backtest Orchestration and Compute Agent

Division: Data and Platform

Schedules Bulletproof BT and research jobs under budgets.

Core responsibilities	Queue, CPU/RAM allocation, deduplication, retries, manifests, hashes, artifact collection, and cost accounting.
Required outputs	Job status; run artifacts; compute ledger.
Permissions and prohibitions	Cannot change parameters after launch or prioritize unapproved work.
Activation triggers	Approved experiment or scheduled job.
Key dependencies	Control Plane; Workers; Artifact Store.
Success measures	Throughput, reproducibility, resource efficiency, and fairness.

49. Deployment and Observability Agent

Division: Data and Platform

Builds, deploys, monitors, and rolls back approved packages.

Core responsibilities	Containers; migrations; health checks; release hashes; logs;
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	metrics; traces; alerts; rollback.
Required outputs	Deployment record; health report; rollback result.
Permissions and prohibitions	Requires signed approval; cannot change strategy or risk config.
Activation triggers	Approved release or incident.
Key dependencies	CI/CD; VM2; Observability Stack.
Success measures	Release reliability, uptime, and rollback success.

50. Security and Access Agent

Division: Data and Platform

Enforces least privilege, secrets safety, and agent boundaries.

Core responsibilities	Policies; secret rotation; access audit; unauthorized-action detection; host boundaries; credential scopes.
Required outputs	Access decision; audit report; security incident.
Permissions and prohibitions	Cannot expose secrets to research agents or broaden permissions without approval.
Activation triggers	Agent creation, permission change, scheduled audit, incident.
Key dependencies	Control Plane; Vault; Human Security Owner.
Success measures	No critical secret exposure or unauthorized privilege expansion.

51. Research Registry and Knowledge Graph Agent

Division: Governance and Memory

Maintains canonical institutional memory and relationships.

Core responsibilities	Record hypotheses, trials, features, datasets, results, reviews, decisions, related ideas, failures, and lineage; support similarity and contradiction search.
Required outputs	Immutable records; relationship graph; retrieval index; trial counts.
Permissions and prohibitions	Cannot delete negative results or alter historical records without controlled correction event.
Activation triggers	Every research action and conclusion.
Key dependencies	PostgreSQL/pgvector; Artifact Store.
Success measures	Completeness, retrieval usefulness, and zero orphaned promoted models.

52. Post-Trade Attribution and Firm Performance Agent

Division: Governance and Memory

Explains firm outcomes and feeds production learning back into research.

Core responsibilities	Attribute P&L by strategy, asset, direction, signal, regime, execution, fees, funding, beta, and risk; compare live to expected; prepare daily/weekly/monthly reviews.
Required outputs	Performance report; live-degradation report; research feedback; retirement triggers.
Permissions and prohibitions	Cannot alter books, approve capital, or hide negative attribution.
Activation triggers	Daily close, weekly review, drawdown, and model review.
Key dependencies	Ledger; Portfolio; Execution; Registry.
Success measures	Accurate attribution, actionable degradation diagnosis, and learning-loop closure.

10. Permissions, Governance, and Decision Rights

10.1 Permission classes

Class	Rights
A - Read-only research	Read approved datasets, knowledge, and records; propose artifacts; no canonical writes except proposals.
B - Research execution	Submit approved jobs in isolated workspaces; write immutable artifacts; no production or holdout access.
C - Independent validation	Rerun, audit, reject, and flag; no editing of original research or live capital approval.
D - Production operations	Deploy signed packages and operate approved systems; no strategy or limit changes.
E - Risk control	Reduce, cancel, halt, and quarantine; never increase limits or initiate discretionary positions.

10.2 Human decision rights

- Founder/CIO: research priorities, approved markets, investment philosophy, live promotion, capital allocation, and scaling.
- Head of Quantitative Research: research standards, experiment approval, and research-program governance.
- Independent Risk Owner: firm-level limits, pause decisions, and authority to stop trading.
- Technology Owner: production architecture, release standards, security, and reliability.
- Compliance/Operations Owner: accounts, records, external rules, legal and operational obligations.

10.3 Deterministic versus agentic boundary

- Agentic: research retrieval, hypothesis generation, experiment design, code drafting, interpretation, critique, anomaly investigation, documentation, and coordination.

- Deterministic: position arithmetic, order generation, duplicate prevention, maximum loss, leverage, prop rules, kill switches, balance checks, reconciliation, approval verification, and secret enforcement.

10.4 Required separation of duties

- No proposer approves its own experiment or promotion.
- No execution agent changes approved logic or risk limits.
- No deployment agent approves its own release.
- No risk agent increases exposure.
- No research agent sees detailed sealed-holdout diagnostics.
- No production worker simultaneously has unrestricted source, database, secrets, exchange, and risk-config access.

11. Data, Metadata, and Experiment Contracts

11.1 Required source metadata

source_id, title, author, year, document_type, subject, chapter, section, page, keywords, mathematical_topics, financial_topics, markets, time_horizon, methodology, assumptions, evidence_type, citation

11.2 Required experiment metadata

experiment_id, hypothesis_id, research_family, repository_commit, dataset_version, instrument_universe, timeframe, date_range, features, target, model_or_rule, parameters, fees, slippage, delay, number_of_trials, validation_method, result, robustness_status, rejection_reason, created_at, agent_identity, approval_id

11.3 Required label contract

- Label definition and exact decision meaning.
- Label start and label_end_time.
- Prediction timestamp and feature availability timestamp.
- Overlap, purging, embargo, and sample-weighting policy.
- Costs, barriers, horizon, censoring, and missing-data policy.
- Universe and eligibility rules at each point in time.

11.4 Required immutable lineage

HYPOTHESIS -> DATA SNAPSHOT -> FEATURE VERSION -> EXPERIMENT MANIFEST
-> CODE COMMIT -> ENGINE CONFIG -> RUN OUTPUT -> VALIDATION REPORTS
-> APPROVAL -> CONTAINER HASH -> LIVE DECISIONS -> ORDERS -> FILLS -> ATTRIBUTION

11.5 Primary database domains

research_sources, research_documents, research_chunks, research_citations, research_hypotheses, research_experiments, research_trials, research_results, research_reviews, research_decisions, research_relationships, feature_definitions, data_snapshots, validation_splits, candidate_signals, promotion_decisions, shadow_predictions, live_degradation, agent_actions, incidents, orders, fills, positions

12. Promotion State Machine and Stage Gates

PROPOSED -> SCREENED -> REPLICATED -> ROBUST -> SHADOW -> MICRO_LIVE
-> PORTFOLIO_ELIGIBLE -> PRODUCTION -> DEGRADED -> RETIRED

State	Minimum gate
PROPOSED	Registered hypothesis, rationale, target, null, failure conditions, trial family, and approved research budget.
SCREENED	Minimum effect evidence, no obvious leakage, viable data, plausible cost envelope, and no dependence on a single configuration.
REPLICATED	Independent reproduction and evidence across periods, instruments, definitions, or markets as appropriate.
ROBUST	Trial-aware statistics, adversarial review, stress survival, baseline comparison, and documented limitations.
SHADOW	Production-parity features and decisions; prospective data, cost, fill, latency, and calibration evidence.
MICRO_LIVE	Minimal-capital implementation validation; correct order lifecycle, risk, reconciliation, and acceptable cost divergence.
PORTFOLIO_ELIGIBLE	Positive marginal contribution, capacity, manageable correlation, risk fit, and approved allocation range.
PRODUCTION	Human CIO and Risk approval, signed deployment artifact, monitoring, rollback, and retirement criteria.
DEGRADED	Threshold breach in calibration, cost, performance, drift, risk, or operations; allocation reduced or paused.
RETIRED	Evidence no longer supports use, implementation is uneconomic, or portfolio value is superseded; full history retained.

12.1 Sealed holdout behavior

The Explorer receives only a pass/fail gate and coarse failure category from sealed historical holdouts. Detailed years, instruments, and parameter diagnostics remain unavailable to prevent repeated adaptation to the holdout.

```
{
  "passed": false,
  "failure_category": "temporal_instability"
}
```

13. Deployment Topology

13.1 VM1 - research development

- Document ingestion, retrieval, vector indexing, experiment schemas, feature/alpha DSL, Bulletproof BT adapters, research tests, Senior Quantitative Research Specialist, Hypothesis Architect, Experiment Designer, ML Research, Statistical Reviewer, and Adversarial Auditor.
- Isolated agent workspaces and non-production credentials.
- Primary development and experimentation environment.

13.2 VM2 - durable control and production

- Research Intelligence API, PostgreSQL/pgvector, trial registry, task orchestration, artifact registry, production knowledge service, scheduled jobs, shadow service, portfolio services, execution, deterministic risk, reconciliation, observability, and production containers.
- Production secrets are scoped by service and unavailable to research profiles.
- Immutable records and durable backups reside here.

13.3 Mac - founder command and review

- Research chat, dashboard, literature upload, hypothesis review, approval workflow, decision journal, performance review, and optional Telegram control interface.
- No requirement for persistent production hosting.
- Used to direct, inspect, approve, and review the firm rather than execute unattended trading.

13.4 Three clocks of live operation

- Fast clock - every bar/tick: compute registered features, score approved signals, monitor execution and drift; no research changes.
- Medium clock - daily/weekly: score prospective predictions, update calibration, evaluate costs, and detect distribution shifts.
- Slow clock - weekly/monthly: generate hypotheses, retrain approved models, run validation, and promote only through governance.

14. Phased Delivery Roadmap

Phase	Primary delivery	Exit condition
RI-1 Research Knowledge Foundation	Ingestion, parsing, metadata, PostgreSQL/pgvector, hybrid search, source retrieval, citation preservation, retrieval evaluation.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
RI-2 Senior Quantitative Research Specialist	Cited research synthesis, prior-experiment inspection, structured hypothesis generation, methods, assumptions, and failure conditions.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
RI-3 Trial Registry and Experiment Contracts	Global trial IDs, immutable manifests, data/code lineage, family counts, negative results, and selection-aware records.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
RI-4 Closed Five-Agent Research Loop	Senior Specialist, Experiment Specification, Execution, Statistical Reviewer, and Adversarial Auditor with human approval.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
ADL-1 Feature/Alpha DSL and Market API	Constrained symbolic language, causal compiler, point-in-time API, query logging, and sealed holdouts.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
ADL-2 Autonomous Factor Miner	Explorer, search scheduler, successive halving, bounded candidate generation, and interpretability.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
ADL-3 Adversarial Research League	Market adversary, red team, falsifier, governor, and trial-aware discovery score.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
ML-1 Reality-Aligned Labels and	Barrier labels, label_end_time,	Acceptance tests passed; artifacts

Phase	Primary delivery	Exit condition
Baselines	logistic/linear baselines, calibrated meta-labelling, cost and fill models.	registered; human owner signs phase completion.
PV-1 Signal Library and Portfolio Validation	Forecast normalization, portfolio contribution, covariance, capacity, turnover, and prop overlay.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
SH-1 Shadow Market Service	Prospective features, predictions, hypothetical orders, observable fills, latency, and live degradation.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
LIVE-1 Micro-Live and Deterministic Operations	OMS, pre-trade risk, real-time risk, reconciliation, incident response, and minimal-capital validation.	Acceptance tests passed; artifacts registered; human owner signs phase completion.
SCALE-1 Controlled Production	Allocation governance, scaling criteria, continuous model risk, attribution, retirement, and feedback into research.	Acceptance tests passed; artifacts registered; human owner signs phase completion.

14.1 First working demonstration

23. Upload a statistics text, time-series text, selected papers, and Invariance experiment reports.
24. Ask one quantitative research question.
25. Retrieve exact relevant passages and prior experiment records.
26. Generate one cited, structured hypothesis.
27. Check the proposal against similar and failed Invariance experiments.
28. Generate a locked experiment specification with costs, validation, and failure conditions.
29. Require human approval before Bulletproof BT execution.
30. Return the result to independent statistical and adversarial review.
31. Register the complete trial regardless of outcome.

15. Metrics and Acceptance Criteria

15.1 Research system metrics

- Percentage of hypotheses falsified cheaply before full backtesting.
- Time from research question to registered experiment.
- Replication rate and trial-adjusted survivor rate.
- Number of genuinely independent signal families.
- Redundant or repeated research avoided through institutional memory.
- Value of information per compute unit.
- Ratio of negative to positive results retained and retrievable.

15.2 Validation metrics

- Critical leakage issues found before promotion.
- Reproduction success rate.
- Calibration of statistical confidence grades versus prospective performance.
- Correlation between stress results and live degradation.
- Percentage of complex models that beat transparent baselines net of costs.

15.3 Portfolio and production metrics

- Marginal portfolio contribution, drawdown, diversification, turnover, capacity, and cost survival.
- Predicted versus realized volatility, correlation, cost, fill, and slippage.

- Shadow-to-live parity and live degradation decomposition.
- Operational error rate, duplicate-order rate, reconciliation breaks, uptime, and kill-switch latency.
- Prop-firm pass probability, breach probability, expected account lifetime, and expected payout before breach.

15.4 Agent quality metrics

- Tasks completed without human correction.
- Output-schema compliance and evidence traceability.
- Unauthorized action attempts blocked.
- Duplicate work avoided.
- Useful findings per tool call and compute unit.
- Escalation quality and reviewer acceptance rate.
- Failure to disclose negative evidence or conflicts: target zero.

15.5 Product acceptance criteria

- Every promoted model has complete immutable lineage.
- No candidate can skip the state machine.
- Every autonomous session has an objective, trial budget, compute budget, allowed tools, and stop condition.
- No research agent can access sealed holdout details or production secrets.
- All live orders require deterministic pre-trade approval.
- Positions, orders, fills, balances, and risk are independently reconciled.
- Every live strategy has warning, reduction, pause, rollback, and retirement criteria.
- Negative results remain searchable and influence future proposals.
- Live degradation is measured and classified before scaling.

16. Risks, Failure Modes, and Guardrails

Risk	Required guardrail
Automated p-hacking	Trial registry, family counts, sealed holdouts, FDR, DSR, PBO, compute budgets, and almost no reward for in-sample performance.
Leakage and timestamp artifacts	Point-in-time API, availability metadata, label_end_time, purging, embargo, delayed features, and independent audit.
Simulator exploitation	Constrained DSL, immutable engine, red-team testing, alternate bar construction, and production parity.
One-agent conflict of interest	Separation among proposer, designer, executor, reviewer, falsifier, governor, deployer, and risk owner.
Research memory loss	Canonical registry, knowledge graph, artifact lineage, and mandatory recording of failures.
Model complexity creep	Transparent baselines, complexity penalty, incremental-value gates, and model cards.
Live self-modification	Three-clock architecture; live anomaly scouts can propose but cannot trade new discoveries.
Prop-rule overfitting	Separate overlay and provider-specific configuration; alpha remains provider-agnostic.
Operational loss	Deterministic risk, reconciliation, scoped secrets, observability, rollback, backups, and incident playbooks.
False confidence from synthetic markets	Synthetic results are robustness evidence only; historical,

Risk	Required guardrail
	shadow, and micro-live evidence remain mandatory.
Over-centralized language model	Research intelligence belongs to shared services and databases so models can be replaced without losing memory.
Human rubber-stamping	Decision memos expose dissent, unresolved risks, evidence grades, and explicit approval accountability.

17. Appendices

Appendix A - Standard hypothesis object

```

hypothesis_id: HYP-2026-0017
title: Volatility-conditioned intraday continuation
research_family: momentum
status: proposed
research_question: Does directional continuation increase after volatility expansion
  when order-flow imbalance and cross-sectional strength agree?
rationale:
  mathematical: [conditional dependence, volatility clustering, state-dependent returns]
  market: [delayed information incorporation, liquidity-taking continuation]
sources:
  - {source_id: PAPER-0012, relevant_section: Section 4}
  - {source_id: BOOK-0043, relevant_section: Chapter 9}
universe: {market: crypto_perpetuals, instruments: liquid_usdt_perpetuals, timeframe: 5m}
features: [realized_volatility, cross_sectional_return_rank, volume_imbalance, funding_rate,
  spread_proxy]
target: {type: forward_return, horizonBars: 12}
null_hypothesis: Conditional forward return is not different from zero after costs.
validation: [expanding_walk_forward, purged_cv, embargo, bootstrap, multiple_testing_adjustment,
  fee_stress, slippage_stress, delay_stress, regime_stability]
failure_conditions: [cost_failure, one_instrument_only, narrow_parameter_peak, oos_failure,
  infeasible_turnover]
priority: medium
estimated_compute: moderate

```

Appendix B - Standard autonomous research mandate

```

Objective: Investigate whether volatility expansion modifies short-term continuation
  across liquid crypto perpetuals.
Limits:
  max_hypotheses: 20
  max_model_specifications: 100
  fixed_compute_budget: true
  live_deployment: prohibited
  promotion_without_approval: prohibited
Required outputs:
  registered hypotheses, full trial ledger, screening results, validation results,
  negative results, adversarial findings, and final research conclusion

```

Appendix C - Pre-deployment checklist

- Complete hypothesis, experiment, data, feature, model, and code lineage.
- Independent reproduction completed.
- Leakage and bias audit resolved.
- Trial-aware statistics and adversarial validation complete.
- Portfolio contribution and capacity approved.
- Shadow evidence demonstrates production parity and acceptable prospective behavior.
- Micro-live plan, risk limits, kill switches, reconciliation, monitoring, and rollback tested.

- Human CIO, Risk, and Technology approvals recorded.
- Warning, reduction, pause, degradation, and retirement criteria configured.

Appendix D - Foundational doctrine

What the firm can guarantee

The firm cannot guarantee profitable alpha. It can build a structural advantage in search, falsification, execution learning, portfolio combination, and adaptation. That process advantage is the product.

What the firm is building

Not an agent that always finds an edge, but a closed autonomous research institution in which one agent proposes, another formalizes, another executes, another attacks, another controls selection bias, another observes live behavior, another governs promotion, every action is recorded, and no agent can alter the rules used to judge itself.

END OF PRODUCT REQUIREMENTS DOCUMENT